

FARUKH MUMTAZ

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11 May 2026

International Exchange Office

Middle East Technical University (METU), Ankara, Türkiye

To the Selection Committee,

SUBJECT: APPLICATION FOR SEMESTER EXCHANGE PROGRAM — FALL 2026

I am a **4th-semester BS Software Engineering student** at COMSATS University Islamabad, Wah Campus, maintaining a CGPA of **3.46/4.0**, and I am applying for the **Fall 2026 Semester Exchange Program** at METU. At this point in my degree, I have done enough independent work to understand where my technical foundation is genuinely weak — and I know that fixing those gaps takes more than self-study. I need to be in an environment with higher academic expectations, stronger peers, and a curriculum that demands rigour. METU fits that description well, and studying there for a semester is the most practical step I can take right now toward becoming a more complete engineer.

Some of the projects I have worked on include **Aegis-Flow**, a PostgreSQL-backed API where I tried to build access-control hardening into the design from the start — which turned out to be harder than expected and exposed real gaps in my systems knowledge. **AWS Security Guardian** is a bot that monitors IAM roles across AWS environments and triggers automated remediation when permissions drift; building it made clear how much I was operating on surface-level cloud knowledge rather than a solid architectural understanding. **LLM Security Gateway** is middleware for blocking prompt injection on AI inference pipelines — I built it because the problem was real and the existing tooling was thin. Each of these taught me something, but each one also showed me what I still cannot do confidently.

Three courses at METU address those gaps directly. **Algorithms (CNG315)** will help me formalize the complexity reasoning I already use in security work but have never had to make rigorous — I apply it intuitively, and I want to be able to back it up properly. **Computer Organization (CNG331)** will give me a clearer picture of how software behaviour connects to hardware, which matters for understanding low-level vulnerabilities at a level beyond intuition. **Software Design (SNG330)** will push me to think about system architecture before jumping into implementation — something my project experience has taught me to value, but that I have not been formally trained in. These three courses sit at the foundation of the kind of engineering I want to do, and I have not covered them with the depth I need.

The perspective I develop at METU will carry directly into my final year project at COMSATS. I want the work I submit in my last semester to be grounded in principle, defensible under scrutiny, and built to a standard that goes beyond what I could produce without this experience. That is the outcome I am working toward, and this exchange is the most direct path to it.

I am available for an interview at your convenience and welcome any questions from the committee. Thank you for your time and consideration.

Sincerely,

Farukh Mumtaz

FA24-BSE-005

COMSATS University Islamabad, Wah Campus